Sharing Energy Data in Data Rooms

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Abstract

Energy consumption data in logistics properties remains fragmented across organizational silos, hindering effective comparison and benchmarking. Existing centralized platforms require stakeholders to relinquish control of their data control, which conflicts with privacy requirements and commercial interests. We present the GRANERGIZE WebApp: A Solid-based application that enables decentralized sharing of building energy data while preserving data sovereignty. The application uses Web Access Control for fine-grained permissions and the Solid Application Interoperability specification for access notifications. In addition to sharing individual building data, the system supports the creation and sharing of aggregated views that only expose computed statistics without revealing underlying buildings. Our implementation demonstrates how the Solid protocol can solve real-world data-sharing issues in the logistics sector, allowing collaboration without centralized data custody.

Keywords

Solid, Linked Data, Access Control,

1. Introduction

Energy consumption data in logistics properties remains trapped in isolated silos, hindering access and exchange. The logistics ecosystem's complexity – with its diverse stakeholders and varied data formats – creates heterogeneity that prevents effective data sharing. Consequently, comparing energy consumption data across properties and organizations remains rudimentary at best.

Logistics properties form central hubs of logistics networks and therefore play a decisive role in the provision of logistics services (source: Nehm et al.)

Quelle? Nehm, A., Veres-Homm, U. and Kille, C. (2009). Logistikimmobilien in Deutschland: Markt und Standorte; eine Studie mit der Unterstützung von Deka Immobilien, Goldbeck, ING Real Estate, Jones Lang LaSalle, ProLogis. Fraunhofer-Verlag, Stuttgart.

. With the rapid rise in electricity and gas prices [1]

Für die direkt darauffolgende Quelle vom Statistischen Bundesamt gibt es mittlerweile eine aktualisierte Version von 2026:
Statistisches Bundesamt (30.01.2026). Statistischer Bericht –
Daten zur Energiepreisentwicklung – Dezember 2025. URL:
https://www.destatis.de/DE/Themen/Wirtschaft/Preise/Publikationen/Energiepreise/statistischer-bericht-energiepreisentwicklung-5619001.html [02.02.2026]

and the increasing importance of sustainability and CO2 accounting through specific EU taxonomies [2], the relevance of energy efficiency in logistics real estate is becoming ever clearer.

Yet substantial barriers persist: The stakeholder landscape's complexity – with landlords, tenants, and operators holding different perspectives on energy use – limits access to the data that is essential for efficient management. Even within organizations, mobilizing consumption data requires extensive manual effort. Furthermore, the diversity of building types and usage patterns makes meaningful comparison difficult.

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Table 1
Overview of stakeholder data exchange pairs involving the facility manager.

Data flow	Data points	Granularity	Periodicity
Facility Manager → Broker	Consumption data (for energy performance certificate)	Consumption types (electricity, water, gas)	annually
Broker → Facility Manager	User for new properties (as sales support)	Total property	once
Facility Manager → Tenant	Consumption data	Consumption types (electricity, water, gas), as granular as possible	At least monthly, ideally every 15 minutes via dashboard
	Blind loads[1]	Total property	once
	Peak loads	Types of consumption: electricity (lighting, intralogistics, etc.)	As detailed as possible[2]
Tenant → Facility Manager	Consumption data (for properties not under management by FM)	Total consumption per user	monthly
	Building specifics (area, age, shift regime, etc.)	Number, size, location	annually

Existing centralized platforms fail to address these barriers because they require stakeholders to relinquish data control and adopt singular data models. The Solid protocol, with its emphasis on decentralized data ownership and standardized access control, offers an alternative: stakeholders retain sovereignty over their data while enabling controlled sharing. This paradigm shift aligns with both GDPR principles and the commercial realities of competitive logistics markets.

To address these challenges, the GRANERGIZE project has developed an ontology that provides a unified data model for energy consumption data, as well as a WebApp for visualizing and sharing this data, which is stored on the user's Solid Pods. The WebApp provides visualizations of the energy consumption data and allows users to define access rights for real or aggregated data via an intuitive user interface. Whereas aggregated data is computed based on the user's preferences and shared as a snapshot without provenance information, individual building data can be shared directly from the owner's Pod with fine-grained access control.

This paper describes the Solid-based architecture ....

describe contributions and maybe sections overview

2. Scenario

The energy consumption of logistics properties is relevant for a wide range of stakeholders. While users, owners, and facility managers are primarily interested in specific consumption data for reporting and calculation purposes, developers, financiers, brokers, energy suppliers, and benchmark service providers rely on data for property comparisons, accurate energy reporting, and consumption optimization.

In order to record the respective interests and data requirements in a structured manner, the project surveyed each stakeholder group about the implications of energy consumption data for their daily business, the associated pains and gains, and any data requirements they might have from other stakeholders. It was assumed that data would only be made available to external stakeholders if they could in turn cover their own data requirements.

Three pairs of stakeholders are listed here as examples, each of which has complementary data requirements and could complement each other well and with low barriers in a suitably designed data room (see Table 1 for the data room involving the facility manager).

[1]@@@link to Wikipedia? link to ontology?

Table 2
Overview of stakeholder data exchange pairs between investor and benchmark service provider. The benchmark service provider takes the shared data points (building specifics and consumption data) as basis for the calculated consumption per m^2, which is in turn shared back.

Data flow	Data points	Granularity	Periodicity
Investor $\rightarrow$ Benchmark service provider	Building specifics (area, age, shift regime, etc.)	Total property	once
	Consumption data	Consumption types (electricity, water, gas)	annually
Benchmark service provider $\rightarrow$ Investor	Min/max/median consumption per m^2 per property	Consumption types (electricity, water, gas)	annually

This list can be supplemented by at least nine other identified pairs, each of which has expressed mutual data requirements in the context of energy consumption or property data for logistics real estate.

In any case, it is necessary to exchange data beyond the boundaries of one's own company, which, due to the data silos that have been common up to now, different data formats and granularities, and general reservations about sharing data, is only possible with a great deal of individual effort.

This is to be addressed and resolved through collaborative data sharing using the GRANERGIZE web app developed as part of the project.

2.1. Data Room

The term Data Room was chosen deliberately, as these Data Rooms are conceptualised as demarcated areas with clearly defined access conditions. Access to such a room is controlled by a gatekeeper. The Data Room therefore represents a protected and restricted area in which only approved participants are permitted to enter on the basis of a corresponding authorisation.

In the scenario considered here, the facility manager (FM) acts as the owner of his Data Room. Various tenants and brokers are granted access to this room in order to find out what data the FM provides at which level of granularity and with what periodicity. If the data offered by the FM meets their requirements, the actors involved can submit their own data offers in return and agree to exchange data if there is mutual consent. If, on the other hand, the available data does not meet the visitors' expectations, they can leave the Data Room at any time without further interaction.

3. Implementation

In the following, we describe the implementation of the GRANERGIZE WebApp. Following the Solid principles of separating identity, data, and application, the WebApp operates as a pure client-side application that communicates directly with Solid Pods without requiring a central backend. All user data – building metadata, energy measurements, and sharing configurations – resides exclusively in the users' personal Solid Pods.

3.1. System Architecture

The GRANERGIZE WebApp is implemented as a single-page application using React 18 with Vite as the build tool and Material-UI for the user interface components. Users authenticate via the Solid-OIDC protocol, which allows them to log in with their WebID and connect to the Identity Provider of their choice. All RDF processing occurs client-side using the N3.js library, which provides parsing and serialization of Turtle documents as well as an in-memory quad store for pattern matching.

might drop this subsection entirely if space is tight

[2] @@@how is that a period? peak load needs to be measured precisely when it happened, precise timestamp, difference between when measured and when communicated; two timestamps in a way, measure timestamp is relevant, you can communicate every month

3.2. Data Model

To maximize interoperability, the GRANERGIZE data model builds upon established ontologies and introduces custom terms only where necessary. For example, the data model reuses the vcard ontology for postal addresses, the RealEstateCore (REC) ontology for building classifications, and Schema.org for organizational entities. Additionally, geographic regions – cities and administrative districts – are identified using Wikidata IRIs, enabling links to external knowledge graphs. The GRANERGIZE vocabulary (`gra:`) defines domain-specific vocabulary for logistics buildings and energy measurements.

Static information on each building, such as: type, geographic coordinates, postal address, total area, and year of construction is stored as a separate Turtle file within the user's Pod, see https://granergize-homer.solidcommunity.net/private/granergize/buildings/2.ttl for an example. Additionally, the building files contain optional properties capturing domain-specific information such as photovoltaic system availability, NACE industry code, or energy certificate references. A building's energy measurement data is stored in separate datasets per year, linked to from the building file. Separating these datasets allows sharing building metadata independently of detailed energy data, supporting fine-grained access control as required by the data providers.

A building's energy data captures the full energy flow within a logistics property using a comprehensive ontology. An example file, containing the detailed energy measurements is available at https://granergize-homer.solidcommunity.net/private/granergize/energy/2023/2.ttl. The model distinguishes six categories: energy need, energy generation, energy storage, energy distribution, energy transfer, and energy usage. Each category contains granular measurement types, such as electricity consumption, photovoltaic generation, heat pump output.

3.3. Data Sharing and Access Control

Access control relies on the Web Access Control (WAC) specification [3], which is part of the Solid platform. WAC associates each resource with an Access Control List (ACL) stored in a corresponding `.acl` file. For defining access rules, ACLs provide a standardized vocabulary (`acl:agent`, `acl:accessTo`, `acl:mode`) to specify read-write permissions on web resources within a Solid Pod. When the data provider grants access to a building, the application appends a new `acl:Authorization` block specifying the data requester's WebID and the granted access mode (typically `acl:Read`). The resource owner retains `acl:Control` permission, allowing them to modify or revoke access at any time. If the data provider chooses to share energy data as well, a separate authorization is added to the energy data file's ACL, enabling fine-grained control over which data elements are accessible.

handle overflows when all sections complete

The sharing workflow follows the Solid Application Interoperability (INTEROP) specification [4] for notifying data requesters about granted access. Figure 1 illustrates the sharing process. The data provider initiates sharing through the web interface by selecting a building and entering the recipient's WebID. Upon confirmation, the application executes three operations on the data provider's Pod (steps 1–3 in the figure): first, it updates the resource's ACL file (`building.ttl.acl`) to grant read permission to the recipient; second, it records the sharing action in `sharingRegistry.ttl`; third, it posts an `interop:AccessGrant` notification to the recipient's inbox. The notification specifies the granter's WebID (`interop:grantedBy`), a timestamp (`interop:grantedAt`), and a data grant pointing to the shared resource.

When the data requester's application polls the inbox (steps 4–5), it discovers the new access grant and automatically adds the shared building URI to `dataSources.ttl`. From this point on, the data requester can fetch the building data directly from the data provider's Pod (step 6). The ACL on Pod A ensures that only authorized requesters – those whose WebID appears in the ACL – can successfully retrieve the resource.

The application maintains local registries to track sharing relationships on both sides. The data provider's `sharingRegistry.ttl` contains triples of the form <buildingUri> gra:sharedWith <recipientWebId>, enabling the WebApp to display which buildings have been shared and allowing users to easily revoke access. On the data requester's side, `dataSources.ttl` lists all external building

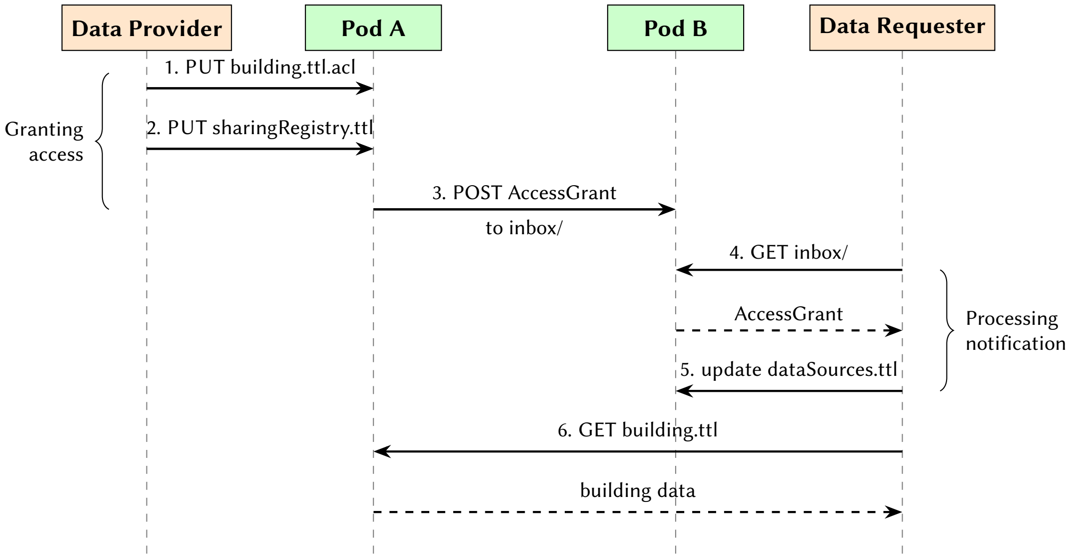

Figure 1: Sequence diagram of the data sharing process: the data provider grants access by updating the ACL and sending a notification; the data requester processes the notification and can then fetch the shared data.

URIs the user has been granted access to. Together, these registries ensure that both parties maintain a consistent view of their data sharing relationships without requiring a central coordination service.

Revocation mirrors the granting process by reversing each step. The application removes the recipient's `acl:Authorization` block from the resource's ACL file and updates `sharingRegistry.ttl` to delete the sharing record. On the data requester's side, subsequent access attempts return HTTP 403, prompting the application to remove the inaccessible building from `dataSources.ttl`.

3.4. Sharing Aggregated Data

While sharing individual building data enables direct collaboration, many scenarios require sharing only statistical summaries without exposing the underlying building details. For instance, a logistics provider managing multiple warehouses may wish to share total energy consumption across their portfolio with an industry benchmarking initiative, without revealing individual building locations or identities. The GRANERGIZE WebApp addresses this requirement through aggregated views – snapshots that only contain computed values.

as described in 2. scenario?

An aggregated view is defined by three parameters: a set of source buildings, an aggregation function (`sum`, `average`, `minimum`, or `maximum`), and a selection of energy metrics to include. Figure 2 shows a screenshot of the user interface for creating an aggregated view. In this screenshot the user selected three buildings and the `average` function for gas and electricity metrics. The view definition is stored as a separate Turtle file in the owner's Pod, as shown in Listing 1. Crucially, the view definition itself – including the source building URIs – is never shared. Only the computed snapshot, containing aggregated values without provenance information, is made accessible to recipients.

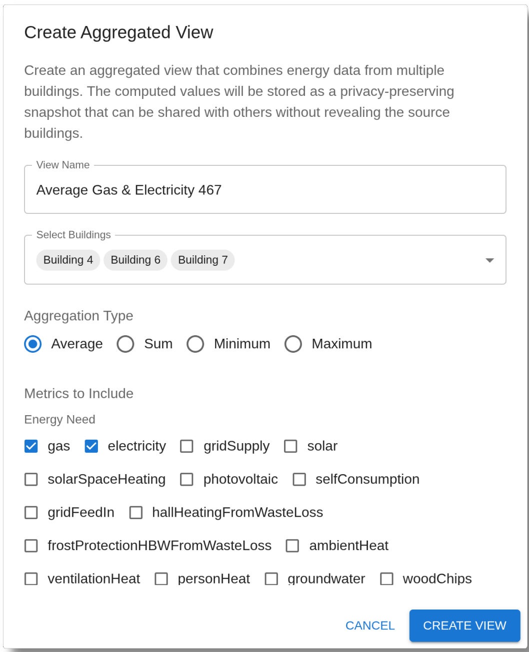

Figure 2: Create Aggregated View Dialog

```
<#view-1770048602886-f5n4sea> a gra:AggregatedViewDefinition ;
  gra:viewId "view-1770048602886-f5n4sea" ;
  gra:viewName "Average Gas and Electricity 467" ;
  gra:aggregationType "average" ;
  gra:createdAt "2026-02-02T16:10:02.886Z"^^xsd:dateTime ;
  gra:includesBuilding <../private/granergize/buildings/4.ttl>,
    <../private/granergize/buildings/2.ttl>,
    <../private/granergize/buildings/6.ttl> ;
  gra:includesMetric "gas", "electricity" ;
  gra:lastComputedAt "2026-02-02T16:10:04.064Z"^^xsd:dateTime .
```

Listing 1: View Definition Example

The aggregation workflow proceeds as follows. When the data owner creates or refreshes a view, the application fetches energy data from each source building, applies the selected aggregation function across all buildings for each metric, and stores the result as a separate Turtle file (the snapshot).

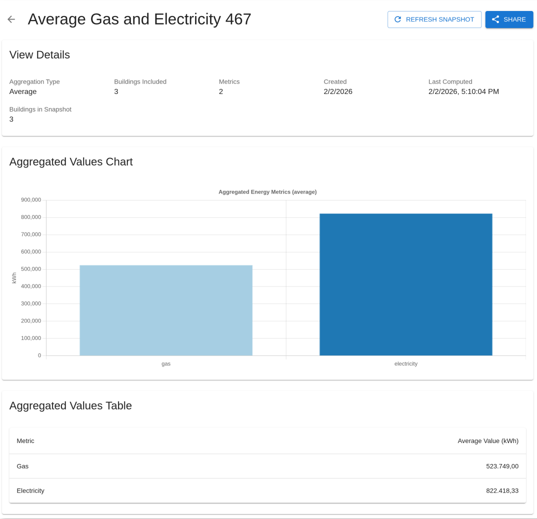

Figure 3: Create Aggregated View Dialog

```
<#snapshot> a gra:AggregatedViewSnapshot ;
  gra:viewId "view-1770048602886-f5n4sea" ;
  gra:viewName "Average Gas and Electricity 467" ;
  gra:aggregationType "average" ;
  gra:computedAt "2026-02-02T16:10:04.064Z"^^xsd:dateTime ;
  gra:buildingCount 3 ;
  gra:includesMetric "gas", "electricity" ;
  gra:gasValue 523749.00 ;
  gra:electricityValue 822418.33 .
```

Listing 2: View Definition Example

Figure 3 shows an example of such a snapshot for the view defined in Listing 1. The snapshot contains only the aggregated values and metadata such as the computation timestamp, building count, and time period – no individual building identifiers appear in the output. This materialized snapshot approach ensures that recipients cannot infer which specific buildings contributed to the aggregation.

Sharing an aggregated view follows the same access control pattern as individual buildings. The application updates the snapshot's ACL file to grant read permission to the recipient's WebID and posts an access notification to their inbox.

View owners retain full control over their aggregated data. Because snapshots are materialized rather than computed on-demand, the shared values reflect the state at the time of creation or last refresh.

Owners can update the snapshot at any time by triggering a manual refresh, which recomputes the aggregation from current source data and overwrites the previous snapshot. Recipients automatically see the updated values on their next fetch, while the owner maintains the ability to review changes before publishing. Access revocation works identically to individual building sharing: removing the recipient's authorization from the ACL immediately prevents further access.

4. Conclusion

Energy consumption data in the logistics sector remains fragmented across organizational boundaries, preventing effective benchmarking and collaboration. Centralized platforms require stakeholders to surrender data control, conflicting with privacy requirements and competitive interests.

This paper presented the GRANERGIZE WebApp, a Solid-based application that addresses these challenges through decentralized data sharing while preserving data sovereignty. The implementation demonstrates how the Solid protocol's core components – Web Access Control for fine-grained permissions and the Application Interoperability specification for access notifications – can solve real-world data-sharing problems. By separating identity, data, and application concerns, the system enables collaboration without centralized data custody. The aggregated views feature extends the sharing model to support statistical summaries, allowing stakeholders to contribute to benchmarking initiatives without exposing individual building details.

Future work includes integrating automated access request workflows, data integration pipelines, and supporting additional aggregation functions such as weighted averages based on building size. We also plan to evaluate the system's usability with logistics stakeholders in a field study to identify practical adoption barriers.

5. Acknowledgements

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References

[1] Federal Statistical Office of Germany (Statistisches Bundesamt), Data on Energy Price Trends: Long-time Series from January 2005 to December 2022, Technical Report, Federal Statistical Office of Germany (Statistisches Bundesamt), 2023.
[2] European Commission, Eu taxonomy for sustainable activities, 2022. URL: https://finance.ec.europa.eu/sustainable-finance/tools-and-standards/eu-taxonomy-sustainable-activities_en.
[3] T. Berners-Lee, H. Story, S. Capadisli, Web Access Control, W3C Draft Report, W3C, 2024. Https://solidproject.org/TR/wac/.
[4] J. Bingham, E. Prud'hommeaux, E. Pavlik, Solid Application Interoperability, W3C Draft Report, W3C, 2025. Https://solid.github.io/data-interoperability-panel/specification/.